
Wave 3 — Trigger, Concurrency, and Public API Safety

Objective

Prevent index-trigger churn during long or bulk workflows without breaking current trigger behavior.

Findings Resolved

- EXH-009
- Supports EXH-001/EXH-002 safety by reducing overlapping workflow risk

Files Affected

- Master_List/Current Production Script/v.1.6.29_Current_Production_Script

Functions Affected

- handleSpreadsheetChangeForIndex
- formatMonthlySheets
- updateDemoPMonthlySync
- createDisenrolledList
- Template rebuild functions
- Trigger setup helpers

Exact Proposed Changes

1. Add document-property workflow busy flag:
 - Set at start of destructive/bulk workflows.
 - Clear in finally.
 - Include timestamp/TTL for recovery.
2. Modify handleSpreadsheetChangeForIndex():

- Keep current document lock and signature logic.
 - Skip or defer index rebuild while busy flag is active.
3. Add logging for skipped/deferred index rebuild.

Business-Logic Impact

None expected; Index may refresh after workflow instead of during workflow.

Breaking-Change Risk

Low.

Data-Integrity Risk

Low; primarily operational/concurrency safety.

Required Tests

- Trigger index rebuild during monthly formatting.
- Bulk sheet create/delete operations.
- Two-user overlap simulation.
- Busy flag stale timestamp recovery.

Recommended Version Increment

v1.6.30 if bundled with Wave 1, otherwise v1.6.29.1.

Independent?

Yes, but more valuable if implemented before/with destructive workflow remediation.

Required User Decisions

- Should busy operations skip index refresh entirely or mark a deferred refresh to run at workflow completion?